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Fault Detection, Analysis, and Rectification Report

Control Systems of Technological Equipment, their Reliability and Diagnostics
FESTO Processing Station Diagnostics

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1. Introduction

This report provides a diagnostic and rectification guide for the FESTO Processing Station. The aim of the lab work was to identify important failure situations, detect them using PLC logic, and provide clear lamp indications so that an operator or maintenance engineer can quickly find the possible cause.

The station performs a basic automated process: a part is detected in the socket, moved to the drilling position, drilled, checked by the testing cylinder, and then moved to the final position. The added diagnostic logic monitors this process using sensor feedback, timers, flags, and the three available indicator lamps.

The system uses the following diagnostic methods:

- sensor inputs for material and position confirmation,
- timer-based detection for movement timeouts,
- logical checks for impossible sensor states,
- lamp indications for operator feedback,
- reset logic for fault recovery.

2. Executive Summary

Five fault categories were implemented for the FESTO Processing Station. They cover the main process, mechanical, pneumatic, electrical, and safety-related problems that can occur during operation.

No.	Fault Category	Root Cause Domain	Indication
1	No part detected at start	Process / material	O1.0 blinking
2	Test socket position fault	Mechanical / electrical / sensor	O1.0 + O1.1 blinking
3	Drill cylinder fault	Pneumatic / mechanical / sensor	O1.0 + O1.2 blinking
4	Testing cylinder fault	Pneumatic / mechanical / sensor	O1.1 + O1.2 blinking
5	Stop or emergency stop condition	Operator / safety	O1.1 blinking

3. Station Description

The FESTO Processing Station consists of a part socket, rotating test socket/table, drilling mechanism, testing cylinder, and final part position. The automatic sequence starts when a part is placed in the socket and the Start button is pressed. The part is moved to the drilling position, drilled, checked by the testing cylinder, and transferred to the final position.

Label	Station Part	Function
A	Part socket	Detects the inserted part
B	Drilling position	Holds the part for drilling
C	Drill	Performs the drilling operation
D	Testing cylinder	Checks the processed part
E	Final slot	Receives the processed part

4. Allocation List

4.1. Inputs

Address	Description
I0.0	Part in socket
I0.1	Test socket position sensor
I0.2	Hold cylinder home position
I0.3	Hold cylinder final position
I0.4	Drill cylinder upper position
I0.5	Drill cylinder lower position
I0.6	Testing cylinder upper position
I0.7	Testing cylinder lower position
I2.0	Start button
I2.1	Reset / clear diagnostics
I2.2	Positioning button
I2.3	Auto / manual control
I2.4	Stop button
I2.7	Emergency stop reset

4.2. Outputs

Address	Description
O0.1	Drill engine

O0.2	Test socket engine
O0.3	Drill down command
O0.4	Drill up command
O0.5	Hold piston home command
O0.6	Testing piston down command
O0.7	Piston holder
O1.0	Start lamp / diagnostic LED 1
O1.1	Reset lamp / diagnostic LED 2
O1.2	Positioning lamp / diagnostic LED 3

4.3. Added Diagnostic Flags and Timers

Address	Description
F0.0	Fault 1: no part detected
F0.1	Fault 2: test socket position fault
F0.2	Fault 3: drill cylinder fault
F0.3	Fault 4: testing cylinder fault
F0.5	Fault 5: stop or emergency stop condition
F0.6	Diagnostic blink OFF phase
F0.7	Diagnostic blink ON phase
T7	Diagnostic LED ON timer
T8	Diagnostic LED OFF timer
T9	Movement fault timeout timer

5. Reset Procedure

The Reset button I2.1 clears active fault flags, blinking flags, diagnostic lamps, and diagnostic timers. After the fault is removed, the operator can reset the indication and return the station to normal operation.

```
IF I2.1 'Reset / clear diagnostics
THEN RESET F0.0
     RESET F0.1
     RESET F0.2
     RESET F0.3
     RESET F0.5
     RESET F0.6
     RESET F0.7
     RESET O1.0
     RESET O1.1
```

```
RESET O1.2
RESET T7
RESET T8
RESET T9
```

6. Detailed Fault Analysis

6.1. Fault 1: No Part Detected at Start

Root Cause Domain: Process / material

Severity: High **Detection Type:** Instantaneous

6.1.1. Fault Condition

This fault occurs when the operator presses Start I2.0, but the part sensor I0.0 does not detect a part in the socket. The cycle must not start because the station would attempt to move, drill, and test without material.

6.1.2. Detection Mechanism

Parameter	Details
Input checked	I0.0 part in socket
Trigger	I2.0 Start button pressed
Fault condition	I2.0 ON and I0.0 OFF
Diagnostic flag	F0.0
Indicator	O1.0 blinking

6.1.3. Possible Causes and Rectification

- Verify that a part is present in the socket.
- Check if the part is seated correctly.
- Inspect and clean sensor I0.0.
- Check the sensor LED, wiring, and connector.
- Replace or adjust the sensor if it does not switch.
- Press Reset I2.1 after correction and start again.

6.1.4. Code Implementation

```
IF I2.0
   AND I2.7
   AND N I0.0
THEN SET F0.0
     RESET F0.1
     RESET F0.2
     RESET F0.3
     RESET F0.5


IF I2.0
   AND I2.7
   AND I0.0
THEN SET P2
```

6.2. Fault 2: Test Socket Position Fault

Root Cause Domain: Mechanical / electrical / sensor

Severity: High **Detection Type:** Timer-based

6.2.1. Fault Condition

This fault occurs when the test socket engine O0.2 is switched ON, but the position sensor I0.1 does not confirm the correct position within the timeout period. If this position is not confirmed, drilling or testing may be misaligned.

6.2.2. Detection Mechanism

Parameter	Details
Output command	O0.2 test socket engine
Sensor feedback	I0.1 test socket position sensor
Timer used	T9 movement fault timeout
Timeout value	5 seconds
Fault condition	T9 finished and I0.1 still OFF
Diagnostic flag	F0.1
Indicator	O1.0 + O1.1 blinking

6.2.3. Possible Causes and Rectification

- Check if the table or test socket physically rotates.

- Inspect the table path for obstruction or mechanical debris.
- Check air supply or motor/actuator power depending on the mechanism.
- Verify that output O0.2 switches correctly.
- Inspect sensor I0.1 alignment and LED indication.
- Check PLC input wiring and connector for I0.1.
- Press Reset I2.1 after the fault is corrected.

6.2.4. Code Implementation

```
IF N T1
THEN SET O0.2
     SET T9 WITH 5s


IF N T9
   AND N I0.1
THEN SET F0.1
     RESET O0.2
     JMP TO 200
```

6.3. Fault 3: Drill Cylinder Fault

Root Cause Domain: Pneumatic / mechanical / sensor

Severity: Critical **Detection Type:** Timer-based and logical conflict

6.3.1. Fault Condition

This fault occurs when the drill cylinder does not reach the commanded upper or lower position within the allowed time. It is also detected if both drill position sensors I0.4 and I0.5 are active at the same time, which is physically impossible.

6.3.2. Detection Mechanism

Parameter	Details
Down command	O0.3 drill down
Up command	O0.4 drill up
Lower sensor	I0.5 drill lower position
Upper sensor	I0.4 drill upper position
Timer used	T9 movement fault timeout

Timeout value	3 seconds
Fault condition	Commanded position not reached, or I0.4 and I0.5 both ON
Diagnostic flag	F0.2
Indicator	O1.0 + O1.2 blinking

6.3.3. Possible Causes and Rectification

- Check pneumatic pressure and air supply to the drill cylinder.
- Inspect the drill cylinder for mechanical blockage or damage.
- Verify solenoid valve operation for O0.3 and O0.4.
- Check sensors I0.4 and I0.5 for alignment and LED state.
- Confirm that only one drill position sensor is active at a time.
- Inspect sensor wiring for loose wires or short circuits.
- Press Reset I2.1 after the fault is removed.

6.3.4. Code Implementation

```
IF O0.1
THEN SET O0.3
     SET T3 WITH 2s
     SET T4 WITH 1s
     SET T9 WITH 3s


IF N T9
   AND N I0.5
THEN SET F0.2
     RESET O0.1
     RESET O0.3
     JMP TO 200


IF N T9
   AND N I0.4
THEN SET F0.2
     RESET O0.1
     RESET O0.4
     JMP TO 200
```

```
IF I0.4
   AND I0.5
THEN SET F0.2
```

6.4. Fault 4: Testing Cylinder Fault

Root Cause Domain: Pneumatic / mechanical / sensor

Severity: High **Detection Type:** Timer-based and logical conflict

6.4.1. Fault Condition

This fault occurs when the testing cylinder is commanded to move down using O0.6, but the lower position sensor I0.7 does not become active within the allowed time. A second fault condition is when upper sensor I0.6 and lower sensor I0.7 are active simultaneously.

6.4.2. Detection Mechanism

Parameter	Details
Output command	O0.6 testing piston down
Upper sensor	I0.6 testing cylinder upper position
Lower sensor	I0.7 testing cylinder lower position
Timer used	T9 movement fault timeout
Timeout value	3 seconds
Fault condition	I0.7 not reached, or I0.6 and I0.7 both ON
Diagnostic flag	F0.3
Indicator	O1.1 + O1.2 blinking

6.4.3. Possible Causes and Rectification

- Check the testing cylinder for free movement.
- Inspect the testing probe for obstruction or incorrect part position.
- Check pneumatic pressure, tubing, and solenoid valve.
- Verify sensor I0.7 lower position detection.
- Verify sensor I0.6 upper position detection.
- Inspect wiring if both sensors are active together.
- Press Reset I2.1 after correction.

6.4.4. Code Implementation

```
IF N T2
THEN SET O0.6
     SET T4 WITH 1s
     SET T9 WITH 3s


IF N T9
   AND N I0.7
THEN SET F0.3
     RESET O0.6
     JMP TO 200


IF I0.6
   AND I0.7
THEN SET F0.3
```

6.5. Fault 5: Stop or Emergency Stop Condition

Root Cause Domain: Operator / safety

Severity: Critical **Detection Type:** Instantaneous

6.5.1. Fault Condition

This fault occurs when Stop I2.4 is pressed or when the emergency stop reset input I2.7 is not active. This condition requires the station to stop and remain stopped until the operator checks the reason.

6.5.2. Detection Mechanism

Parameter	Details
Stop input	I2.4
Emergency reset input	I2.7
Detection type	Instantaneous
Diagnostic flag	F0.5
Indicator	O1.1 blinking

6.5.3. Possible Causes and Rectification

- Identify why Stop or emergency stop was activated.

- Remove any obstruction or unsafe condition.
- Confirm that all motion has stopped.
- Reset the emergency stop circuit.
- Press Reset I2.1 to clear diagnostics.
- Restart only after the station is safe.

6.5.4. Code Implementation

```
IF I2.4
THEN SET F0.4
     SET F0.5


IF N I2.7
THEN SET F0.5
     RESET O0.1
     RESET O0.2
     RESET O0.3
     RESET O0.4
     RESET O0.5
     RESET O0.6
     RESET O0.7
```

7. Lamp Indication Summary

The three available lamps are reused as diagnostic indicators. The blinking pattern identifies the active fault and directs the operator to the first area to inspect.

No.	Fault Title	Lamp Pattern	First Inspection Area
1	No part detected at start	O1.0 blinking	Part socket, I0.0 sensor
2	Test socket position fault	O1.0 + O1.1 blinking	Table movement, I0.1 sensor
3	Drill cylinder fault	O1.0 + O1.2 blinking	Drill cylinder, I0.4/I0.5 sensors
4	Testing cylinder fault	O1.1 + O1.2 blinking	Testing cylinder, I0.6/I0.7 sensors
5	Stop or emergency stop	O1.1 blinking	Stop button, emergency circuit

8. Diagnostic Stop State

When a diagnostic fault occurs during the automatic cycle, the program jumps to Step 200. This step removes the main motion commands before the operator resets or restarts the station.

```
STEP 200
IF NOP
THEN RESET O0.1
     RESET O0.2
     RESET O0.3
     RESET O0.4
     RESET O0.6
     RESET T9
     JMP TO 201
```

9. Testing and Validation

The diagnostic functions were checked by creating each fault condition and observing the diagnostic flag, lamp indication, and program reaction. The expected result is that the correct lamp pattern appears and the unsafe output commands are removed.

Test	Scenario	Expected Result
1	Press Start with no part in socket	F0.0 active, O1.0 blinks, cycle does not start
2	Block I0.1 during table movement	F0.1 active, O1.0 + O1.1 blink, movement stops
3	Block I0.5 during drill down	F0.2 active, O1.0 + O1.2 blink, drill command stops
4	Block I0.4 during drill return	F0.2 active, O1.0 + O1.2 blink, drill up command stops
5	Block I0.7 during testing	F0.3 active, O1.1 + O1.2 blink, testing command stops
6	Force I0.4 and I0.5 ON together	F0.2 active as drill sensor conflict
7	Force I0.6 and I0.7 ON together	F0.3 active as testing sensor conflict
8	Press Stop during automatic cycle	F0.5 active, O1.1 blinks, outputs stop

9	Activate emergency stop condition	F0.5 active, O1.1 blinks, all main outputs reset

10. Conclusion

This report presents a structured diagnostic guide for the FESTO Processing Station. Five important faults were implemented: missing part, test socket position fault, drill cylinder fault, testing cylinder fault, and stop/emergency stop condition.

The diagnostic system uses PLC inputs, outputs, timers, flags, and lamp indications to identify faults and guide troubleshooting. Each fault includes the detection condition, possible causes, and rectification steps, so the report can be used as a practical manual for operators and maintenance engineers.

The final implementation improves fault detection, reduces troubleshooting time, and supports safer operation of the FESTO station.